

Self-assessment questions on Forming by Plastic Deformation

- Indicate when hot forming is preferred in the industrial shaping of metals and alloys
- Explain which phenomena are exploited in hot forming and which are the relevant affecting factors
- Clear up the role played by the strain rate sensitivity coefficient
- Correlate the change of microstructure and mechanical properties during recrystallization of a TMCP
- Discuss why recrystallization is convenient during hot rolling, especially if it is retarded to low temperature
- Prove why the final grain size after recrystallization decreases with increasing straining
- Elucidate when hot forging is favored over hot rolling in terms of product and mechanical properties requirements
- Compare the residual stress consequences resulting from small and large rolls during hot rolling of plates
- Compare hot vs. cold rolling in terms of thickness reduction, surface finish and final mechanical properties
- Underline the factors/performances controlled by normal and planar anisotropy in sheet forming